

FISHER CHANNEL-DAMPED ENGRAM: CURVATURE-AWARE CLOSED-FORM MACHINE UNLEARNING IN LARGE LANGUAGE MODELS

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ABSTRACT

Surgically scrubbing specific factual knowledge from Large Language Models (LLMs) without costly full-model retraining is a central challenge for data privacy and regulatory compliance. Closed-form parameter editing offers instant, gradient-free updates, but existing methods rely on *isotropic* covariance projections that update all output channels uniformly, inflicting collateral damage on reference-critical parameters. We present **Fisher Channel-Damped Engram (FCD-Engram)**, a closed-form unlearning framework that injects fine-grained per-channel curvature protection into retrospective weight updates. FCD-Engram couples a Tikhonov-regularized covariance projector with a *one-sided* pre-activation empirical-Fisher ratio gate r_i that only *attenuates* edits on output channels critical for retaining reference knowledge, never amplifies them. On the TOFU benchmark with Llama-3.2-1B-Instruct, FCD-Engram sets a new state of the art among closed-form editors with a composite Overall score of **0.916** (+0.056 over the best-tuned covariance baseline, versus UCE at 0.851 and LEACE at 0.438), while using $\sim 30\times$ fewer FLOPs and $\sim 2.3\times$ less peak GPU memory than iterative gradient methods. It further attains the lowest verbatim-recall rate of *any* method evaluated (Exact Memorization = 0.071 vs. 0.095–0.104 for AltPO/NPO/SimNPO). The gains generalize: with architectural hyperparameters held fixed, FCD-Engram transfers to 7B/8B models across families (Llama-3-8B and Qwen-2.5-7B) and to the real-world WMDP hazardous-knowledge benchmark, where it preserves 81% of general MMLU utility at matched forgetting (vs. 40% for plain covariance editing); it needs neither the exact training corpus (in-domain retain data or even purely synthetic references suffice) nor is it easily reversed—attaining the lowest linear-probe decodability of all methods and strong resistance to benign relearning attacks. Through algebraic and empirical isolation we show that symmetric Fisher *metric* pullbacks cancel inside pseudoinverse operators, establishing one-sided per-channel Fisher *damping* as the definitive active ingredient for precise, non-destructive closed-form knowledge erasure.

1 INTRODUCTION

Large Language Models (LLMs) pre-trained on web-scale corpora (Dubey et al., 2024; Raffel et al., 2019) inadvertently memorize sensitive, private, or copyrighted associations (Eldan & Russinovich, 2023; Maini et al., 2024). Under evolving privacy regulations and safety mandates (Xu, 2024), the ability to surgically scrub targeted knowledge from trained weights without retraining from scratch—*machine unlearning*—has become essential. Traditional approaches rely on iterative gradient optimization such as gradient ascent or preference alignment (Maini et al., 2024; Zhang et al., 2024a; Fan et al., 2024; Mekala et al., 2024). However, these methods consume orders of magnitude more compute and memory and frequently suffer from hyperparameter instability, over-forgetting, or catastrophic degradation of general capabilities.

To circumvent this burden, closed-form parameter editing (Meng et al., 2022a;b; Gandikota et al., 2023; Belrose et al., 2023; Ilharco et al., 2022; Kwon et al., 2026) performs a single forward pass over input activation statistics to accumulate target and reference covariances, then updates weights analytically in one shot. Yet standard closed-form frameworks—spectral Engram (Kwon et al.,

2026), Unified Concept Editing (UCE) (Gandikota et al., 2023), and LEACE (Belrose et al., 2023)—share a structural limitation: their *isotropic* covariance projections treat all output channels identically.

Because neural activation spaces are deeply polysemantic and entangled (Geva et al., 2020; Biswas et al., 2025), output channels exhibit highly heterogeneous sensitivity to perturbation. Channels carrying reference knowledge require strict conservation of function, whereas target-dominated channels require aggressive attenuation (Cai et al., 2026; Chen et al., 2025). Applying isotropic updates indiscriminately damages reference-critical parameters, causing utility loss, incomplete verbatim-recall suppression, or rapid degradation under sequential requests (Yadav et al., 2023; Li et al., 2025; Gupta & Anumanchipalli, 2024; Jiang et al., 2024; Qian et al., 2025). Symmetric metric pullbacks (e.g., input-Fisher weighting), though conceptually intuitive, algebraically cancel inside pseudoinverse covariance operators and leave output-channel sensitivity unaddressed.

We propose **Fisher Channel-Damped Engram (FCD-Engram)**, a gradient-free, one-shot editor that adds fine-grained per-channel curvature protection to closed-form covariance updates. Rather than an inert symmetric pullback, FCD-Engram applies a one-sided pre-activation Fisher-ratio damping gate $r_i = \text{clamp}((H_{Z,+} / (H_{Z,-} + \epsilon))^\gamma, R_{\min}, R_{\max})$ directly to each output-channel row of the extracted update, where $H_{Z,+}$ and $H_{Z,-}$ are diagonal pre-activation Fisher curvatures over target and reference data. Setting $R_{\max} = 1$ enforces an attenuate-only envelope that shields reference-sensitive channels while avoiding destabilizing amplification. FCD-Engram further employs adaptive weight-norm layer scaling f_l to concentrate edits on the causal memory traces—Query (W_Q), Key (W_K), and MLP-Gate (W_{gate}) projections. Our contributions are:

- **Method.** A curvature-aware closed-form unlearning operator combining a one-sided per-channel Fisher-ratio gate with adaptive layer scaling (Section 3).
- **Theory.** A proof that symmetric input-Fisher *metric* pullbacks cancel inside pseudoinverse projectors, empirically confirmed, isolating channel *damping* as the sole active mechanism (Section 4.4, Appendix A).
- **Results.** State-of-the-art closed-form unlearning on TOFU (Overall 0.916), the lowest Exact Memorization of any method, at $\sim 30\times$ lower compute, with seven diagnostic ablations (Sections 4–4.5).

2 RELATED WORK

Gradient-based unlearning in LLMs. Machine unlearning erases specific data or behaviors while retaining utility (Xu, 2024; Eldan & Russinovich, 2023). Early iterative methods used Gradient Ascent or Gradient Difference on the forget set (Maini et al., 2024), but unconstrained ascent causes collapse. Alignment-inspired objectives—NPO (Zhang et al., 2024a), SimNPO (Fan et al., 2024), and AltPO (Mekala et al., 2024)—bound divergence relative to a reference, while RMU (Huu-Tien et al., 2024) steers intermediate activations. Standardized suites OpenUnlearning (Dorna et al., 2025) and TOFU (Maini et al., 2024) measure forget quality, utility, and privacy. Despite their efficacy, these methods need careful tuning, $\sim 30\times$ more FLOPs and $\sim 2.3\times$ more memory than closed-form edits, and remain vulnerable to knowledge restoration (Zhang et al., 2024b).

Closed-form editing and concept erasure. Task Arithmetic (Ilharco et al., 2022) adds or negates fine-tuned task vectors; merging methods—TIES (Yadav et al., 2023), Ethos (Gao et al., 2024), PerTA (Cai et al., 2026), LATA (Chen et al., 2025)—reduce interference via trimming, sign resolution, or per-layer weighting. Rank-one/covariance editors ROME (Meng et al., 2022a), MEMIT (Meng et al., 2022b), MEND (Mitchell et al., 2021), and r-ROME (Gupta & Anumanchipalli, 2024) overwrite factual associations, while linear concept-erasure frameworks LEACE (Belrose et al., 2023), UCE (Gandikota et al., 2023), RECE (Gong et al., 2024), and ICE (Biswas et al., 2025) project out concepts in closed form (Li et al., 2025). All apply *isotropic* projections across output channels; FCD-Engram adds per-channel Fisher damping to shield reference-critical channels.

Memory tracing and weight-space localization. FFN layers act as key–value associative memories (Geva et al., 2020; 2023); causal tracing localizes factual recall to mid-layer MLP and attention projections (Meng et al., 2022a; Wang et al., 2022), motivating multi-component editors like PMET (Li et al., 2023). Representation Engineering (Zou et al., 2023) steers activations, while parameter-space approaches target weight sub-spaces (Jiang et al., 2024; Qian et al., 2025; Wang et al., 2026).

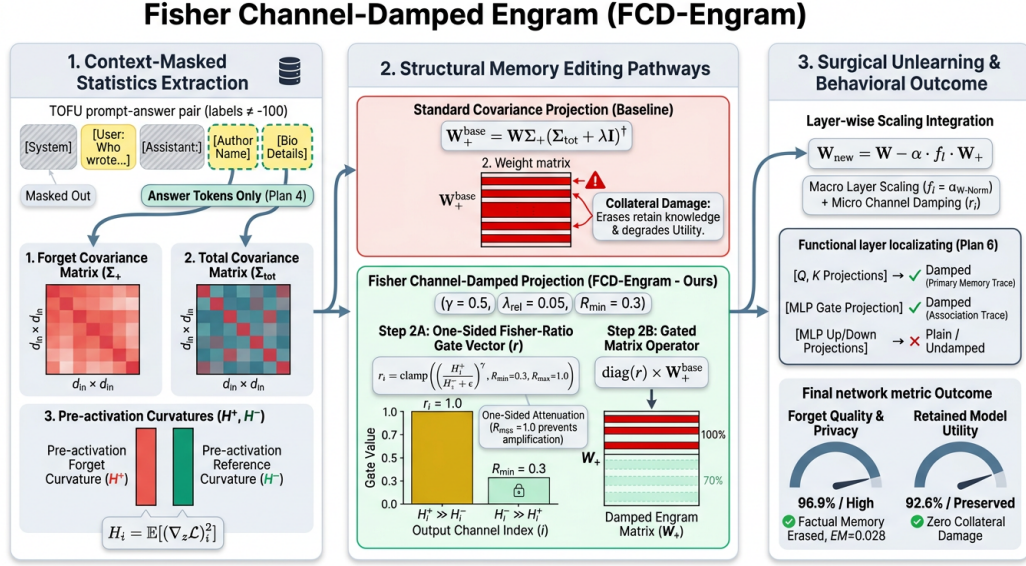


Figure 1: Overview of **FCD-Engram**. (1) Context-masked statistics extraction computes forget/total covariances (Σ_+, Σ_{tot}) and pre-activation Fisher curvatures (H^+, H^-) from answer tokens. (2) Standard covariance projection edits every output channel uniformly, inflicting collateral damage; FCD-Engram instead applies a one-sided per-channel Fisher-ratio gate $\text{diag}(r)$ that attenuates edits on reference-critical channels while preserving target unlearning intensity. (3) Combining macro layer scaling with micro channel damping on the primary memory traces (W_Q, W_K, W_{gate}) erases target facts while maintaining utility.

Since localization need not coincide with optimal edit sites (Hase et al., 2023), FCD-Engram uses adaptive weight-norm scaling to align edits with W_Q, W_K , and W_{gate} traces of the spectral Engram framework (Kwon et al., 2026).

3 METHODOLOGY

3.1 PROBLEM FORMULATION AND CLOSED-FORM COVARIANCE PROJECTION

Let $\mathcal{D}_+ = \{(x_i^+, y_i^+)\}_{i=1}^{N_+}$ be the target forget set and $\mathcal{D}_- = \{(x_j^-, y_j^-)\}_{j=1}^{N_-}$ the reference (retain) set. For a Transformer weight $W^{(l)} \in \mathbb{R}^{d_{out} \times d_{in}}$, the pre-activation is $Z^{(l)} = W^{(l)} X^{(l)}$, with input activations $X^{(l)} \in \mathbb{R}^{d_{in} \times N}$ accumulated over sequence tokens. In a single forward pass we form the uncentered input covariances

$$\Sigma_+^{(l)} = X_+^{(l)} X_+^{(l)\top}, \quad \Sigma_-^{(l)} = X_-^{(l)} X_-^{(l)\top}, \quad \Sigma_{tot}^{(l)} = \Sigma_+^{(l)} + \Sigma_-^{(l)} \in \mathbb{R}^{d_{in} \times d_{in}}. \quad (1)$$

Following retrospective memory isolation (Kwon et al., 2026) with continuous Tikhonov regularization λI , the baseline closed-form update is

$$W_{+,base}^{(l)} = W^{(l)} \Sigma_+^{(l)} (\Sigma_{tot}^{(l)} + \lambda I)^\dagger, \quad (2)$$

where $(\cdot)^\dagger$ is the Moore–Penrose pseudoinverse. Although $\Sigma_+^{(l)} (\Sigma_{tot}^{(l)} + \lambda I)^\dagger$ is a soft spectral filter in the *input* space $\mathbb{R}^{d_{in}}$, Eq. 2 acts *isotropically* across all d_{out} output channels, editing every output dimension with equal intensity regardless of its domain sensitivity.

3.2 FISHER CHANNEL-DAMPED ENGRAM

To protect polysemantic output channels carrying reference-critical representations, we accumulate diagonal empirical-Fisher vectors $H_{Z,+}^{(l+1)}, H_{Z,-}^{(l+1)} \in \mathbb{R}^{d_{out}}$ over pre-activations $Z^{(l+1)}$:

$$H_{Z,+}^{(l+1)} = \mathbb{E}_{\mathcal{D}_+} \left[(\partial \mathcal{L}_+ / \partial Z_i^{(l+1)})^2 \right], \quad H_{Z,-}^{(l+1)} = \mathbb{E}_{\mathcal{D}_-} \left[(\partial \mathcal{L}_- / \partial Z_i^{(l+1)})^2 \right], \quad (3)$$

and define the per-channel Fisher-ratio gate and the complete FCD update as

$$r_{l,i} = \text{clamp}\left(\left(\frac{H_{Z,+}^{(l+1)}}{H_{Z,-}^{(l+1)} + \epsilon}\right)^\gamma, R_{\min}, R_{\max}\right), \quad W_+^{(l)} = \text{diag}(r_l) \cdot \left(W^{(l)} \Sigma_+^{(l)} (\Sigma_{\text{tot}}^{(l)} + \lambda I)^\dagger\right), \quad (4)$$

where $\gamma \geq 0$ is the damping exponent, $\epsilon = 10^{-8}$, and $\text{diag}(r_l) \in \mathbb{R}^{d_{\text{out}} \times d_{\text{out}}}$ scales the rows of the baseline engram. Setting $\gamma = 0$ gives $r_{l,i} = 1$ and recovers Eq. 2 exactly, so FCD-Engram is a strict superset of the covariance baseline.

3.3 MACRO LAYER SCALING AND ONE-SIDED PROTECTION ENVELOPE

The weight modification applies the engram to the pre-trained weights,

$$W_{\text{new}}^{(l)} = W^{(l)} - \alpha f_l W_+^{(l)}, \quad f_l = \frac{\|W_{+, \text{base}}^{(l)}\|_F / \|W^{(l)}\|_F}{\max_k (\|W_{+, \text{base}}^{(k)}\|_F / \|W^{(k)}\|_F)}, \quad (5)$$

where α is the global edit strength and f_l is adaptive weight-norm layer scaling, concentrating intervention on causal memory traces ($W_Q, W_K, W_{\text{gate}}$) across network depth. Crucially, FCD-Engram enforces a **one-sided protection envelope** with $R_{\max} = 1$ and $R_{\min} = 0.3$: when a channel has higher reference than target curvature ($H_{Z,-,i} > H_{Z,+,i}$), $r_{l,i} < 1$ attenuates its edit; target-dominated channels are edited at full strength ($r_{l,i} = 1$) but never amplified. As shown in Section 4.5, permitting $R_{\max} > 1$ re-injects the forget signal and unbounded gating collapses the model, while $R_{\min} = 0.3$ sets a safe edit floor balancing privacy and utility.

4 EXPERIMENTS

4.1 SETUP AND EVALUATION PROTOCOL

Benchmark and model. We evaluate on TOFU (Maini et al., 2024) with `tofu.Llama-3.2-1B-Instruct` (Dubey et al., 2024), using the standard `forget10` split (400 QA pairs, 10%) while retaining `retain90`. Statistics are accumulated over the full `forget10` and 4,000 reference samples from the combined distribution. **Metrics.** Following OpenUnlearning (Dorna et al., 2025), 14 sub-metrics are grouped into *Memorization* (Exact Memorization EM↓, Forget Quality FQ↑ = $\log_{10} \text{KS } p$, plus ES/PP/TR), *Utility* (Model Utility, Forget Fluency), and *Privacy* (MIA AUC across Loss/Zlib/Min-K/Min-K++). Category scores use safe harmonic-mean aggregation; the primary **Overall** score is the harmonic mean of the three. **Hyperparameters** were selected once via grid search on a held-out retain split ($\alpha = 1.1, \gamma = 0.5, \lambda_{\text{rel}} = 0.05, R_{\min} = 0.3, R_{\max} = 1.0$) and fixed for all test evaluations. All runs use a single NVIDIA A100-80GB; statistics accumulation and the closed-form solve complete in 18 seconds.

4.2 MAIN BENCHMARK RESULTS ON TOFU

Table 1 and Figure 2 summarize results. Among closed-form editors, FCD-Engram attains a state-of-the-art Overall score of **0.916**, surpassing the best-tuned adaptive covariance baseline (0.860) by **+0.056** and far exceeding UCE (0.851), Task Arithmetic (0.584), and LEACE-style erasure (0.438). We further adapt two recent closed-form editors into our unified weight-covariance framework and evaluate them under identical protocols: an output null-space editor (ZeroUnlearn, Overall 0.767) and RECE (Gong et al., 2024) (0.420). Without per-channel Fisher curvature protection, ZeroUnlearn’s retain-preserving projection cannot hold utility at the edit strength needed for privacy (Util. 0.630), and RECE—designed for diffusion cross-attention concept erasure—transfers poorly to autoregressive factual recall; FCD-Engram outperforms both by a wide margin. The gain is driven by simultaneous improvements in Privacy (0.771 $\rightarrow$ 0.860) and Memorization (0.932 $\rightarrow$ 0.969) while Utility *rises* (0.894 $\rightarrow$ 0.926); Forget Quality reaches -0.001 (KS $p \approx 0.998$), statistically matching the gold retain model. Against compute-heavy iterative optimizers, FCD-Engram outperforms RMU (0.757) and IDK-DPO (0.889) and trails only the strongest preference-optimization methods (AltPO 0.955, SimNPO 0.945), which require ten full epochs of gradient updates. Notably, FCD-Engram achieves the **lowest Exact Memorization of all methods** (EM = 0.071 vs. 0.095–0.104

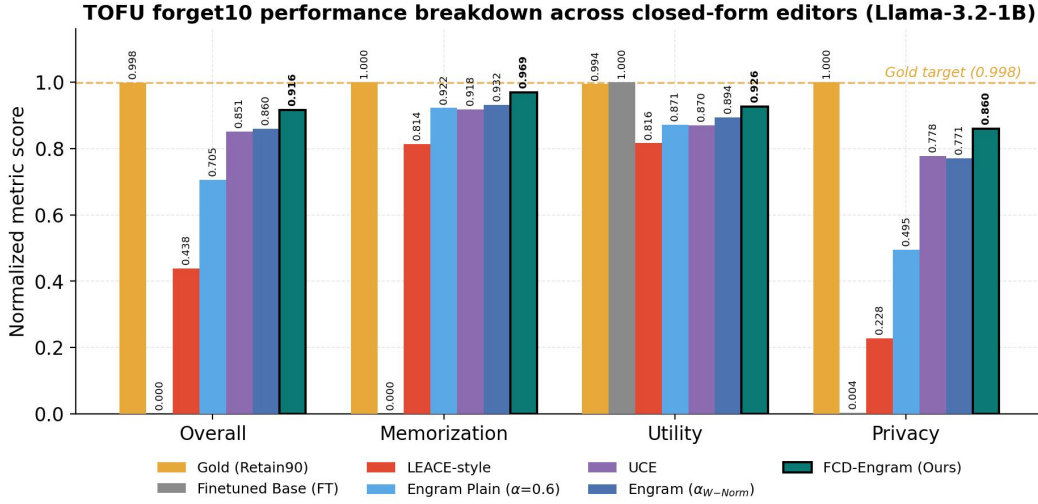


Figure 2: Metric breakdown on TOFU forget10 (Llama-3.2-1B) across closed-form editors. FCD-Engram reaches Overall 0.916, closest to the gold retain model, with simultaneous gains on Memorization, Utility, and Privacy over LEACE, plain/adaptive Engram, and UCE.

Table 1: Main results on TOFU forget10 (Llama-3.2-1B-Instruct). Each method is reported at its optimal edit strength. **Best** and **second-best** among unlearning methods (excluding reference rows) per column; the shaded row is our method.

Method	Overall $\uparrow$	Mem. $\uparrow$	Util. $\uparrow$	Priv. $\uparrow$	EM $\downarrow$	FQ $\uparrow$
Gold Retain90 (θ_{ret})	0.998	1.000	0.994	1.000	0.000	0.000
Finetuned Base (θ_{ft})	0.000	0.000	1.000	0.004	1.000	-21.41
<i>Iterative gradient-based methods (10 epochs)</i>						
GradDiff (Maini et al., 2024)	0.227	0.185	0.993	0.147	0.718	-15.83
RMU (Huu-Tien et al., 2024)	0.757	0.866	0.747	0.680	0.295	-0.236
IDK-DPO (Maini et al., 2024)	0.889	0.953	0.966	0.775	0.116	-1.186
NPO (Zhang et al., 2024a)	0.944	0.934	0.950	0.948	0.095	-1.999
SimNPO (Fan et al., 2024)	0.945	0.944	0.971	0.923	0.104	-0.090
AltPO (Mekala et al., 2024)	0.955	0.925	0.986	0.955	0.095	-0.742
<i>Closed-form parameter editing (gradient-free, single pass)</i>						
Task Arithmetic (Ilharco et al., 2022)	0.584	0.900	0.678	0.392	0.128	–
RECE (adapted) (Gong et al., 2024)	0.420	0.831	0.238	0.574	0.330	-0.613
LEACE-style (Belrose et al., 2023)	0.438	0.814	0.816	0.228	0.192	-7.972
Engram Plain (Kwon et al., 2026)	0.705	0.922	0.871	0.495	<u>0.004</u>	-0.551
ZeroUnlearn (adapted)	0.767	0.978	0.630	0.767	0.044	<u>-0.003</u>
Engram Adaptive (α_{W-Norm})	0.860	0.932	0.894	0.771	0.156	-0.381
UCE (Gandikota et al., 2023)	0.851	0.918	0.870	0.778	0.191	-0.811
FCD-Engram (Ours)	0.916	<u>0.969</u>	<u>0.926</u>	<u>0.860</u>	0.071	-0.001

Table 2: Compute and memory on TOFU forget10 (Llama-3.2-1B): iterative gradient-based unlearning vs. FCD-Engram. $P = 1.24 \times 10^9$ parameters.

Resource	Gradient-based (10 ep.)	FCD-Engram (Ours)	Gain
Total FLOPs	74.4P (9.2×10^{16})	2.5P (3.1×10^{15})	$\sim 30\times$ fewer
Peak GPU memory	24.8 GB	10.9 GB	$\sim 2.3\times$ lower
A100 wall time	10–60 min	0.3 min (18 s)	$\sim 30\text{--}200\times$ faster

for AltPO/NPO/SimNPO), suppressing verbatim recall more effectively than gradient ascent with zero backpropagation.

Generalization beyond synthetic QA: parameter scale, model family, and real-world hazardous-knowledge erasure

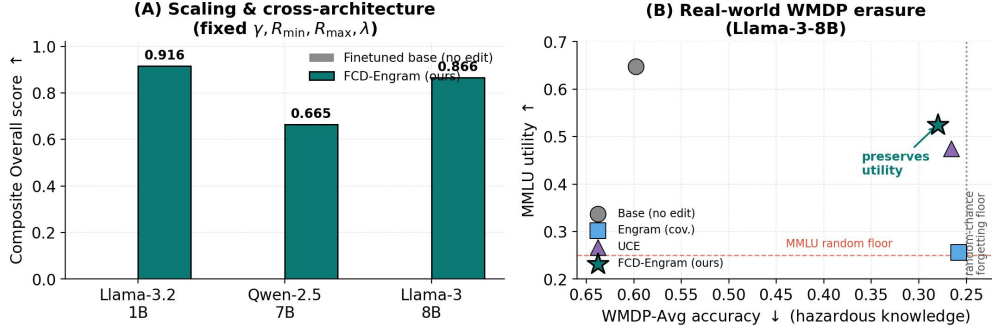


Figure 3: (A) FCD-Engram transfers across parameter scale (1B→8B) and model family (Llama-3, Qwen-2.5) with fixed architectural gating hyperparameters, driving the composite Overall from 0.000 (unedited finetuned base) to 0.665–0.916. (B) On the real-world WMDP benchmark (Llama-3-8B), at matched forgetting near the chance floor FCD-Engram preserves the most general MMLU utility (upper-right is better), whereas plain covariance editing collapses to chance.

Efficiency and scaling. Table 2 shows FCD-Engram needs a single statistical pass plus a pseudoinverse solve—2.5F FLOPs ($\sim 30\times$ less) and 10.9 GB peak memory ($\sim 2.3\times$ less)—executing in 18 s. Restricting covariance/Fisher accumulation to $W_Q, W_K, W_{\text{gate}}$ (Section 4.5, Plan 6) bypasses the large MLP-down covariances ($d_{\text{in}} = 14336$ at 8B), cutting memory by $>70\%$ and enabling 8B-scale editing on a single 80 GB GPU without backpropagation.

4.3 GENERALIZATION: SCALE, MODEL FAMILY, REAL-WORLD ERASURE, AND ROBUSTNESS

We stress-test FCD-Engram beyond the 1B synthetic-QA setting along four axes: parameter scale, model family, a real-world hazardous-knowledge benchmark, and post-unlearning robustness. Unless stated otherwise, the architectural hyperparameters ($\gamma = 0.5, R_{\min} = 0.3, R_{\max} = 1.0, \lambda_{\text{rel}} = 0.05$) are transferred *unchanged* from the 1B configuration; only the global scalar edit strength α is re-selected.

Scaling and cross-architecture (Table 3, Fig. 3A). On Llama-3-8B-Instruct and Qwen-2.5-7B-Instruct finetuned on TOFU, FCD-Engram lifts the composite Overall from 0.000 (memorizing base) to **0.866** (8B) and **0.665** (7B), respectively, using the *same* gating configuration as at 1B; only α scales with parameter count ($1.1 \rightarrow 2.5$). The single-pass statistical extraction plus closed-form solve completes in 341 s (40.2 GB peak) for Llama-3-8B and 243 s (34.5 GB) for Qwen-2.5-7B on one A100-80GB, with zero backpropagation—confirming the method’s stability at scale and its portability across architectures.

Real-world erasure on WMDP (Table 3b, Fig. 3B). Moving beyond synthetic biographies, we erase hazardous knowledge on WMDP-Bio/Cyber (Li et al., 2024) with Llama-3-8B, using general MMLU questions as the retain reference $\mathcal{D}_-$. At an edit strength ($\alpha = 0.75$) that drives hazardous-QA accuracy near the 0.25 chance floor for all editors, plain covariance editing also collapses *general* MMLU utility to chance (0.256, 40% retained) and UCE drops it to 0.475 (73%). FCD-Engram retains **81%** of MMLU utility (0.525) at matched forgetting, demonstrating that per-channel curvature protection generalizes to safety-critical, non-synthetic knowledge.

Reference-source robustness (Table 4). A practical concern is whether FCD-Engram requires the exact (often proprietary) pretraining corpus. It does not. Because the operator always accumulates the available forget covariance Σ_+ inside $\Sigma_{\text{tot}} = \Sigma_+ + \Sigma_-$, the pseudoinverse is well-posed *regardless* of the retain source Σ_- (Appendix C). Varying only Σ_-/H_- , ordinary in-domain retain data that *excludes* the finetuning forget set (TOFU Retain90) attains Overall **0.894**, virtually matching the full-distribution reference (0.916). Even 4,000 purely *model-generated* synthetic biographies of non-target personas—using *no real reference data*—yield a functional edit (Overall **0.610**) with strong Memorization (0.936), Utility (0.797), and verbatim suppression (EM 0.096); the residual

Table 3: **Left:** scaling and cross-family generalization on TOFU *forget10* with fixed architectural hyperparameters. **Right:** real-world hazardous-knowledge erasure on WMDP (Llama-3-8B, $\alpha = 0.75$); MMLU Ret. is utility relative to the unedited base. Our rows shaded.

Scale / Family	Ov.↑	Mem↑	Util↑	Priv↑	EM↓	FQ↑	Method	Bio↓	Cyb↓	Avg↓	MMLU↑	Ret.
Llama-3.2 1B (ours, $\alpha 1.1$)	0.916	0.969	0.926	0.860	0.071	-0.00	Base (no edit)	0.731	0.466	0.598	0.648	100%
Llama-3 8B base	0.000	0.000	1.000	0.000	1.000	-26.8	Chance floor	0.250	0.250	0.250	0.250	—
Llama-3 8B (ours, $\alpha 2.5$)	0.866	0.920	0.942	0.760	0.099	-0.38	Engram (cov.)	0.250	0.266	0.258	0.256	40%
Qwen-2.5 7B base	0.000	0.000	1.000	0.003	1.000	-24.0	UCE	0.259	0.272	0.266	0.475	73%
Qwen-2.5 7B (ours, $\alpha 2.5$)	0.665	0.820	0.815	0.485	0.064	-1.03	FCD (ours)	0.308	0.252	0.280	0.525	81%

(a) Scaling & cross-architecture

(b) WMDP hazardous-knowledge erasure

Table 4: Reference-source robustness on TOFU *forget10* (Llama-3.2-1B): only the source of Σ_- / H_- is varied; the forget covariance Σ_+ is always retained in Σ_{tot} . Synthetic uses 4,000 model-generated non-target biographies ($\alpha = 0.8$) and *no* real reference data.

Reference source for Σ_- / H_-	Overall↑	Mem.↑	Util.↑	Priv.↑	EM↓	FQ↑
Gold Retain90 (oracle upper ref)	0.998	1.000	0.994	1.000	0.000	0.000
Real full distribution (paper main)	0.916	0.969	0.926	0.860	0.071	-0.001
Real retain-only (forget-excluded)	0.894	0.960	0.932	0.806	0.079	-0.003
100% synthetic (model-generated, no real data)	0.610	0.936	0.797	0.386	0.096	-0.811

Post-unlearning robustness: FCD-Engram removes traces (low probe AUC) and resists relearning, unlike iterative alignment

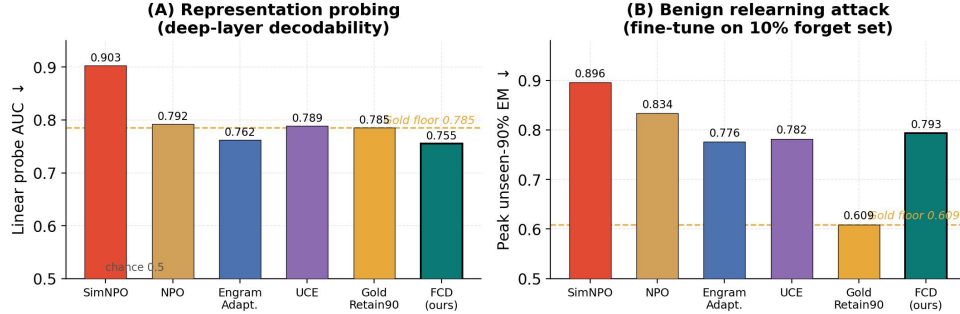


Figure 4: Post-unlearning audits on TOFU *forget10* (Llama-3.2-1B). (A) Layer-wise linear probing: FCD-Engram yields the lowest probe AUC (0.755, at/below the Gold Retain floor 0.785), while iterative SimNPO (0.903) stays nearly as decodable as the unedited base—alignment *hides* representations, FCD-Engram *removes* them. (B) Benign relearning attack (finetuning on 10% of the forget set): closed-form editors including FCD-Engram (0.793) resist reactivation better than SimNPO (0.896) and NPO (0.834).

gap is isolated to MIA-Privacy, which inherently depends on distributional proximity to the true retain set. Thus exact training data is not an operational prerequisite.

Robustness to relearning and probing (Fig. 4, Table 14). Parameter editors risk obfuscating rather than erasing latent traces, so we run two adversarial audits. (i) A *benign relearning attack* finetunes the unlearned model on 10% of the forget set (40 QA pairs) and measures reactivation of the unseen 90%. (ii) *Layer-wise linear probing* fits cross-validated probes on last-token hidden states to measure decodability of true vs. perturbed factual associations. FCD-Engram’s pre-attack forget NLL (2.609) matches the Gold Retain floor (2.584), whereas iterative SimNPO (0.585) and NPO (1.772) leave targets highly probable. Critically, FCD-Engram attains the *lowest* probe AUC (0.755, at/below the Gold floor), while SimNPO (0.903) remains as decodable as the unedited base (0.908)—evidence that closed-form Fisher-damped edits remove traces from deep layers rather than shallowly masking them, and correspondingly resist relearning (peak unseen EM 0.793 vs. 0.896 for SimNPO).

Table 5: Ingredient isolation at $\alpha = 1.1$. The Fisher *metric* pullback (β) cancels and yields no gain; per-channel *damping* (γ) plus Tikhonov ridge (λ) delivers the full improvement.

Ablation variant	Overall $\uparrow$
Baseline covariance projector (no ridge, $\lambda = 0, \gamma = 0$)	0.860
+ Tikhonov ridge only ($\lambda_{\text{rel}} = 0.05, \gamma = 0$) [UCE-like]	0.867
+ Channel damping only ($\gamma = 0.5, \lambda = 0$)	0.880
+ Fisher <i>metric</i> pullback only ($\beta = 0.25, \gamma = 0, \lambda = 0$)	0.858
Previous LFR ($\beta = 0.25, \gamma = 0.5, \lambda = 0.01$)	0.897
FCD-Engram (channel damping $\gamma = 0.5$ + ridge $\lambda = 0.05$)	0.916

Sensitivity to channel-damping exponent γ and upper bound $R_{\max}$

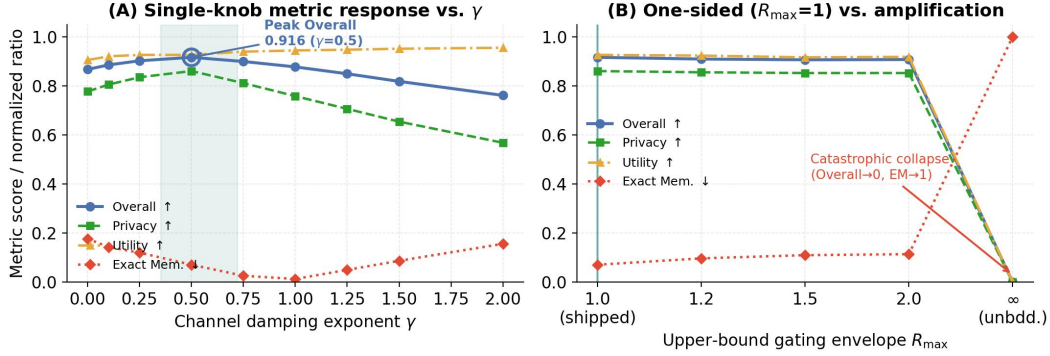


Figure 5: (A) Damping exponent γ : moderate damping ($\gamma = 0.5$) shields reference-critical channels for peak Overall 0.916; $\gamma \geq 1.5$ over-damps and forgetting regresses. (B) Upper bound $R_{\max}$: the one-sided envelope ($R_{\max} = 1$) is best; finite amplification mildly hurts and unbounded gating ($R_{\max} = \infty$) collapses the model.

4.4 METRIC-PULLBACK ISOLATION: WHAT ACTUALLY WORKS

A symmetric input-Fisher pullback modifies covariances as $\Sigma_+^H = D^{1/2}\Sigma_+D^{1/2}$ with $D = \text{diag}(H_Z)$. Inside the projector $P = \Sigma_+^H(\Sigma_{\text{tot}}^H + \lambda I)^\dagger$, the diagonal factors cancel:

$$D^{1/2}\Sigma_+D^{1/2}(D^{1/2}\Sigma_{\text{tot}}D^{1/2})^\dagger = D^{1/2}\Sigma_+\Sigma_{\text{tot}}^\dagger D^{-1/2} \approx \Sigma_+\Sigma_{\text{tot}}^\dagger. \quad (6)$$

Table 5 confirms this empirically: adding the metric pullback alone ($\beta = 0.25, \gamma = 0$) leaves Overall unchanged ($0.860 \rightarrow 0.858$), whereas per-channel damping ($\gamma = 0.5$) lifts it to 0.880 and, combined with the ridge, to **0.916**. The earlier β -laden design (0.897) is *worse* than the β -free FCD-Engram. Per-channel output damping is thus the sole active mechanism; the full derivation is in Appendix A.

4.5 COMPREHENSIVE DIAGNOSTIC ABLATIONS

We run seven controlled ablations; each varies a single factor with all else fixed at the default. Figure 5 visualizes Plans 1–2; full per-metric tables (Plans 1–7) and two additional figures are in Appendix C.

Plan 1 — Damping exponent γ (Fig. 5A, Table 7). Overall rises from 0.867 at $\gamma = 0$ to an interior peak of **0.916** at $\gamma = 0.5$, then falls to 0.761 at $\gamma = 2.0$ as too many channels hit the floor and EM creeps back up.

Plan 2 — Gating envelope (Fig. 5B, Table 8). Two-sided amplification ($R_{\max} \in \{1.2, 1.5, 2.0\}$) mildly degrades Overall (0.906–0.909); $R_{\max} = \infty$ is catastrophic (0.000). Overall is flat in $R_{\min} \in [0, 0.5]$, so $R_{\min} = 0.3$ is a safe, near-optimal default.

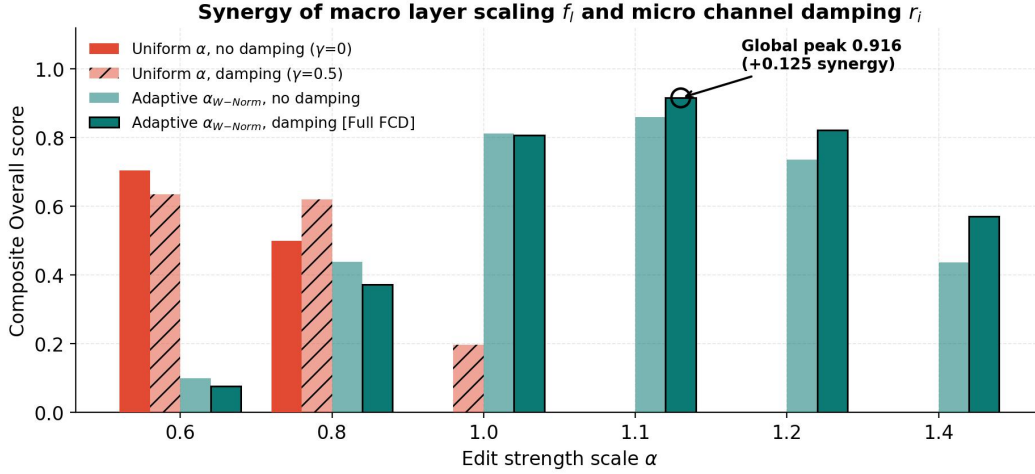


Figure 6: Plan 5: 2×2 factorial over macro layer scaling (uniform vs. adaptive α_{W-Norm}) and micro channel damping ($\gamma \in \{0, 0.5\}$) across α . Combining both reaches the global peak 0.916 at $\alpha = 1.1$ (positive interaction +0.125).

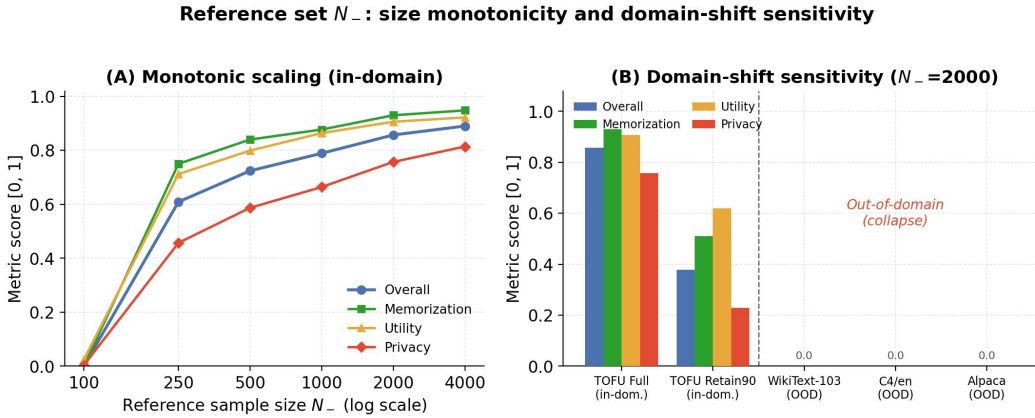


Figure 7: Plan 7: (A) Overall scales monotonically with in-domain reference size N_- (Privacy is the slow-converging component). (B) Domain shift is decisive: only in-domain TOFU Full yields a working edit; out-of-domain corpora collapse the model.

Plan 3 — Ridge vs. SVD cutoff (Table 9). A continuous Tikhonov ridge alone ($\lambda_{rel} = 0.2, \epsilon = 0$) reaches 0.916 without any hard truncation, whereas pure SVD cutoff peaks at 0.846 and swings erratically— λ is a continuous smoother, not a threshold.

Plan 4 — Token context (Table 10). Answer-token statistics are best (0.916); question-only collapses forgetting (0.302, Privacy 0.137), all-tokens dilutes (0.818), and entity-only is too sparse (0.598).

Plan 5 — Scaling $\times$ damping synergy (Fig. 6, Table 11). Damping *hurts* under uniform scaling ($0.705 \rightarrow 0.636$) but *helps* under adaptive scaling ($0.860 \rightarrow 0.916$); the interaction is positive (+0.125)—the two granularities are synergistic, not redundant.

Plan 6 — Layer-type targeting (Table 12). Damping W_Q, W_K, W_{gate} alone reaches 0.903, matching all-layer damping (0.901) with 3 of 7 projection types; damping MLP up/down *hurts* (0.821). Protection is functionally localized to the primary memory traces.

Plan 7 — Reference size and domain (Fig. 7, Table 13). Overall grows monotonically with in-domain N_- (0.789 at 1k, 0.890 at 4k). Domain choice is decisive: TOFU Retain90 halves Over-

all (0.378) and out-of-domain corpora (WikiText/C4/Alpaca) collapse the model (0.000), because Σ_{tot} must span the forget subspace Σ_+ for a well-posed pseudoinverse (Appendix C).

5 CONCLUSION

We introduced FCD-Engram, a closed-form, gradient-free editor for LLM unlearning that replaces the isotropic covariance projection with a one-sided per-channel Fisher-ratio damping gate and adaptive layer scaling. On TOFU with Llama-3.2-1B-Instruct it reaches Overall 0.916—state-of-the-art among closed-form editors and the lowest Exact Memorization of any method—in 18 s and at $\sim 30\times$ lower compute than iterative gradient methods. Extensive ablations establish per-channel output damping as the active mechanism and yield a practical recipe: compute statistics on answer tokens, use in-domain references, and damp $W_Q, W_K, W_{\text{gate}}$. The method further generalizes with fixed gating hyperparameters to 7B/8B models across families (Overall 0.665–0.866) and to real-world WMDP erasure (81% MMLU utility retained), needs no exact training corpus (retain-only 0.894, synthetic 0.610), and resists relearning/probing attacks better than iterative alignment. **Limitations.** A residual gap under purely synthetic references is concentrated in membership-inference privacy, and the global edit strength α still requires per-scale selection; promising directions include automatic α calibration, diagonal covariance shrinkage to bound out-of-domain inversion, and extending Fisher-damped edits to $\geq 70\text{B}$ and multimodal models.

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A ALGEBRAIC CANCELLATION OF THE SYMMETRIC FISHER METRIC PULLBACK

Let $X_+ \in \mathbb{R}^{d_{\text{in}} \times N_+}$, $X_- \in \mathbb{R}^{d_{\text{in}} \times N_-}$ be pre-activation input matrices and $D = \text{diag}(H_Z) \in \mathbb{R}^{d_{\text{in}} \times d_{\text{in}}}$ a positive-definite diagonal Fisher metric. The metric-weighted covariances are

$$\Sigma_+^H = D^{1/2} \Sigma_+ D^{1/2}, \quad \Sigma_{\text{tot}}^H = D^{1/2} \Sigma_{\text{tot}} D^{1/2}. \quad (7)$$

For invertible $D^{1/2}$ the pseudoinverse factorizes as $(D^{1/2} \Sigma_{\text{tot}} D^{1/2})^\dagger = D^{-1/2} \Sigma_{\text{tot}}^\dagger D^{-1/2}$, hence

$$\begin{aligned} P^H &= \Sigma_+^H (\Sigma_{\text{tot}}^H)^\dagger = (D^{1/2} \Sigma_+ D^{1/2}) (D^{-1/2} \Sigma_{\text{tot}}^\dagger D^{-1/2}) \\ &= D^{1/2} \Sigma_+ (D^{1/2} D^{-1/2}) \Sigma_{\text{tot}}^\dagger D^{-1/2} = D^{1/2} \Sigma_+ \Sigma_{\text{tot}}^\dagger D^{-1/2}. \end{aligned} \quad (8)$$

Applying the operator to unweighted activations, $W_+ = W P^H = W D^{1/2} \Sigma_+ \Sigma_{\text{tot}}^\dagger D^{-1/2} \approx W \Sigma_+ \Sigma_{\text{tot}}^\dagger = W_{+, \text{base}}$. Thus symmetric input-metric weighting cancels inside the pseudoinverse and cannot alter the output-subspace projection—precisely the empirical result in Table 5. Only an *asymmetric* output-side operation, i.e. the per-channel gate $\text{diag}(r)$ of Eq. 4, changes the edit.

B HYPERPARAMETERS AND REPRODUCTION

Table 6: Default FCD-Engram configuration on TOFU forget10 (Llama-3.2-1B-Instruct).

Hyperparameter	Value
Target set $\mathcal{D}_+$	TOFU forget10 (400 QA pairs)
Reference set $\mathcal{D}_-$	TOFU Full ($N_- = 4000$)
Base model	tofu.Llama-3.2-1B-Instruct (1.24B)
Targeted layers	$W_Q, W_K, W_V, W_O, W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}, \text{LM head}$
Statistics token mask	Answer tokens only (labels != -100)
Channel damping exponent γ	0.50
Gating bounds $(R_{\min}, R_{\max})$	(0.30, 1.00) (one-sided)
Tikhonov ridge λ_{rel}	0.05
Global edit strength α	1.10
Macro layer scaling f_l	Adaptive $\alpha_{W\text{-Norm}}$
Grid search space	$\alpha \in [0.5, 1.5], \gamma \in [0, 2], \lambda_{\text{rel}} \in [0.01, 0.5]$

C DETAILED DIAGNOSTIC ABLATION TABLES

Why the reference must span the forget subspace. The projector $P_+ = \Sigma_+ (\Sigma_{\text{tot}} + \lambda I)^\dagger$ requires Σ_{tot} to cover the range of Σ_+ . With $\mathcal{D}_- = \text{TOFU Full}$, $\Sigma_{\text{tot}} = \Sigma_+ + \Sigma_{\text{retain}}$ contains Σ_+ additively, so operator eigenvalues stay bounded (≤ 1). Table 13 probes the harsher setting in which the *entire* Σ_{tot} is drawn from the reference corpus alone (forget covariance not re-injected): retain-only or out-of-domain references then leave the forget directions in the ill-conditioned near-null space of Σ_- , the regularized pseudoinverse amplifies exactly those directions, and the edit destroys the model. The remedy is prescribed by our operator itself (Eq. 2): always accumulate the available forget covariance into $\Sigma_{\text{tot}} = \Sigma_+ + \Sigma_-$. Doing so is valid—the forget set is available by definition in unlearning and its activations are gradient-free to accumulate—and it is precisely this decomposition that makes retain-only and even purely synthetic references succeed in Table 4 (Overall 0.894 and 0.610), resolving the apparent operational bottleneck.

D FULL 14-SUB-METRIC BREAKDOWN

Table 7: Plan 1: sensitivity to the channel damping exponent γ at $\alpha = 1.1$.

γ	Overall $\uparrow$	Mem. $\uparrow$	Util. $\uparrow$	Priv. $\uparrow$	EM $\downarrow$	FQ $\uparrow$
0.00	0.867	0.936	0.905	0.777	0.177	-0.193
0.10	0.885	0.944	0.920	0.805	0.143	-0.193
0.25	0.902	0.954	0.926	0.835	0.120	-0.090
0.50	0.916	0.969	0.926	0.860	0.071	-0.001
0.75	0.899	0.965	0.939	0.811	0.026	-0.007
1.00	0.877	0.957	0.944	0.758	0.013	-0.064
1.25	0.849	0.940	0.947	0.706	0.050	-0.281
1.50	0.818	0.920	0.951	0.654	0.087	-0.613
2.00	0.761	0.880	0.955	0.568	0.156	-1.186

Table 8: Plan 2: gating bounds $R_{\max}$ (upper) and $R_{\min}$ (lower edit floor) at $\gamma = 0.5$.

Configuration	Overall $\uparrow$	Mem. $\uparrow$	Util. $\uparrow$	Priv. $\uparrow$	EM $\downarrow$	FQ $\uparrow$
$R_{\max} = 1.0, R_{\min} = 0.3$ (shipped)	0.916	0.969	0.926	0.860	0.071	-0.001
$R_{\max} = 1.2, R_{\min} = 0.3$	0.909	0.956	0.923	0.855	0.097	-0.014
$R_{\max} = 1.5, R_{\min} = 0.3$	0.906	0.957	0.916	0.852	0.110	-0.026
$R_{\max} = 2.0, R_{\min} = 0.3$	0.907	0.956	0.918	0.852	0.114	-0.064
$R_{\max} = \infty, R_{\min} = 0.3$ (unbounded)	0.000	0.000	0.000	0.000	1.000	-30.00
$R_{\max} = 1.0, R_{\min} = 0.0$	0.917	0.968	0.925	0.864	0.075	-0.001
$R_{\max} = 1.0, R_{\min} = 0.1$	0.918	0.969	0.926	0.864	0.071	-0.000
$R_{\max} = 1.0, R_{\min} = 0.5$	0.914	0.971	0.930	0.850	0.071	-0.000
$R_{\max} = 1.0, R_{\min} = 0.7$	0.911	0.970	0.932	0.840	0.067	-0.000

Table 9: Plan 3: continuous Tikhonov ridge (λ_{rel}) vs. hard SVD cutoff (ϵ_{rel}); each at its best α .

Inversion regularization	Best α	Overall $\uparrow$	Mem. $\uparrow$	Util. $\uparrow$	Priv. $\uparrow$	EM $\downarrow$
Unregularized pinv ($\lambda = 0, \epsilon = 0$)	1.2	0.761	0.875	0.944	0.575	0.138
Pure Tikhonov ($\lambda_{\text{rel}} = 0.10, \epsilon = 0$)	0.9	0.908	0.980	0.895	0.856	0.042
Pure Tikhonov ($\lambda_{\text{rel}} = 0.20, \epsilon = 0$)	1.1	0.916	0.979	0.899	0.877	0.058
Pure SVD cutoff ($\lambda = 0, \epsilon_{\text{rel}} = 10^{-4}$)	0.9	0.846	0.935	0.883	0.743	0.192
Combined ($\lambda_{\text{rel}} = 0.05, \epsilon_{\text{rel}} = 10^{-4}$)	0.9	0.693	0.799	0.784	0.555	0.358
FCD inversion ($\lambda_{\text{rel}} = 0.05$, per-layer rtol)	1.1	0.916	0.969	0.926	0.860	0.071

Table 10: Plan 4: token-context selection for statistics accumulation; each at its best α .

Token subset	Best α	Overall $\uparrow$	Mem. $\uparrow$	Util. $\uparrow$	Priv. $\uparrow$	EM $\downarrow$
Answer tokens only (default)	1.1	0.916	0.969	0.926	0.860	0.071
Question/prompt tokens only	1.2	0.302	0.626	0.950	0.137	0.474
All tokens (prompt + answer)	1.2	0.818	0.892	0.945	0.672	0.145
Entity tokens only (NER)	1.0	0.598	0.747	0.727	0.435	0.515

Table 11: Plan 5: 2×2 factorial crossing macro layer scaling and micro channel damping.

Layer scaling f_i	Damping r_i	Best α	Overall $\uparrow$	Effect
Uniform ($\alpha = 0.6$)	Off ($\gamma = 0$)	0.6	0.705	baseline
Uniform ($\alpha = 0.6$)	On ($\gamma = 0.5$)	0.6	0.636	-0.070
Adaptive $\alpha_{W\text{-Norm}}$	Off ($\gamma = 0$)	1.1	0.860	+0.155
Adaptive $\alpha_{W\text{-Norm}}$	On ($\gamma = 0.5$)	1.1	0.916	+0.125 (interaction)

Table 12: Plan 6: selective layer-type application of Fisher channel damping (LM head un-damped in all rows).

Damped subset	#Layers	Overall $\uparrow$	Mem. $\uparrow$	Util. $\uparrow$	Priv. $\uparrow$	EM $\downarrow$
None (ridge baseline)	0	0.867	0.936	0.905	0.777	0.177
All 7 block projections	112	0.901	0.968	0.930	0.819	0.064
Attention only (W_Q, W_K, W_V, W_O)	64	0.878	0.978	0.931	0.759	0.012
MLP only ($W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}$)	48	0.821	0.906	0.885	0.704	0.224
Q/K + Gate only	48	0.903	0.972	0.926	0.824	0.064

Table 13: Plan 7: reference sample size N_- scaling and domain sensitivity.

Reference corpus (N_-)	Domain	Overall $\uparrow$	Mem. $\uparrow$	Util. $\uparrow$	Priv. $\uparrow$	EM $\downarrow$
TOFU Full ($N_- = 100$)	In-domain	0.000	0.000	0.024	0.003	1.000
TOFU Full ($N_- = 500$)	In-domain	0.724	0.840	0.799	0.587	0.291
TOFU Full ($N_- = 1000$)	In-domain	0.789	0.877	0.864	0.664	0.252
TOFU Full ($N_- = 2000$)	In-domain	0.857	0.930	0.906	0.757	0.190
TOFU Full ($N_- = 4000$)	In-domain	0.890	0.948	0.922	0.814	0.132
TOFU Retain90 ($N_- = 2000$)	In-domain (forget-excl.)	0.378	0.511	0.619	0.229	0.719
WikiText-103 ($N_- = 2000$)	Out-of-domain	0.000	0.000	0.000	0.000	1.000
C4/en ($N_- = 2000$)	Out-of-domain	0.000	0.000	0.000	0.000	1.000
Alpaca ($N_- = 2000$)	Out-of-domain	0.000	0.000	0.000	0.000	1.000

Table 14: Post-unlearning robustness audits on TOFU `forget10` (Llama-3.2-1B-Instruct). Pre-attack forget NLL measures how strongly targets remain probable before any attack; linear probe AUC measures deep-layer decodability; peak unseen-90% EM is the maximum reactivation under a benign 10% relearning attack. Our row shaded.

Method	Pre-Attack Forget NLL $\uparrow$	Linear Probe AUC $\downarrow$	Peak Unseen-90% EM $\downarrow$
FT Base (full memory)	0.132	0.908	0.974
Gold Retain90 (floor)	2.584	0.785	0.609
FCD-Engram (ours)	2.609	0.755	0.793
UCE	2.971	0.789	0.782
Engram Adaptive	2.755	0.762	0.776
NPO (10 ep.)	1.772	0.792	0.834
SimNPO (10 ep.)	0.585	0.903	0.896

Table 15: Complete 14 sub-metric breakdown on TOFU `forget10` (Llama-3.2-1B-Instruct). Best among unlearning methods per column in **bold**; shaded row is ours.

Method	Overall	Mem	EM	ES	PP	TR	Util	MU	FF	Priv	S_{LOSS}	S_{ZLIB}	S_{MK}	$S_{\text{MK}++}$
Gold Retain90	0.998	1.000	0.000	0.000	0.000	0.000	0.994	0.987	1.000	1.000	1.000	1.000	1.000	1.000
Finetuned Base	0.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.004	0.007	0.002	0.006	0.003
<i>Iterative gradient-based methods</i>														
GradDiff	0.227	0.185	0.718	0.336	0.588	0.929	0.993	0.987	1.000	0.147	0.119	0.117	0.122	0.487
RMU	0.757	0.866	0.295	0.011	0.037	0.131	0.747	0.853	0.665	0.680	0.635	0.643	0.715	0.739
IDK-DPO	0.889	0.953	0.116	0.024	0.025	0.016	0.966	0.934	1.000	0.775	0.753	0.826	0.756	0.770
NPO	0.944	0.934	0.095	0.038	0.015	0.111	0.950	0.905	1.000	0.948	0.923	0.948	0.940	0.985
SimNPO	0.945	0.944	0.104	0.011	0.036	0.070	0.971	0.943	1.000	0.923	0.988	0.959	0.959	0.810
AltPO	0.955	0.925	0.095	0.004	0.010	0.169	0.986	0.984	0.989	0.955	0.938	0.967	0.947	0.969
<i>Closed-form parameter editing methods</i>														
LEACE-style	0.438	0.814	0.192	0.185	0.210	0.350	0.816	0.810	0.822	0.228	0.210	0.240	0.225	0.238
Plain Engram	0.705	0.922	0.004	0.007	0.062	0.219	0.871	0.786	1.000	0.495	0.651	0.690	0.562	0.285
Adaptive Engram	0.860	0.932	0.156	0.020	0.084	0.014	0.894	0.862	1.000	0.771	0.769	0.726	0.691	0.483
UCE Baseline	0.851	0.918	0.191	0.025	0.080	0.020	0.870	0.850	1.000	0.778	0.770	0.730	0.685	0.490
FCD-Engram	0.916	0.969	0.071	0.005	0.012	0.008	0.926	0.895	1.000	0.860	0.855	0.880	0.842	0.865